PROJECT PROPOSAL

Predicting Disease-Causing Non-Coding DNA Variants

Broad topic: *How can we understand the function of non-coding DNA variants to predict whether they may cause disease?*

This document narrows that topic into three fully-specified, executable computational projects plus one stretch idea. Each follows the same funnel as the worked PPI example (broad topic → sharper question → concrete pipeline with numbered steps).

The narrowing logic (the funnel)

How do we tell if a non-coding variant causes disease?

- **A** — **Candidate A · Variant-to-gene pipeline for one disease**
  Take a disease's GWAS loci and output ranked causal-variant → mechanism → target-gene hypotheses.
- **B** — **Candidate B · Benchmark + consensus of the predictors**
  Do AlphaGenome / CADD / SpliceAI / Borzoi agree, who is right where, and does combining them beat the best one?
- **C** — **Candidate C · Splicing-focused prioritization for a gene panel**
  Find rare deep-intronic / near-splice variants that create cryptic splice sites and are candidate-pathogenic.
- **D** — **Candidate D (stretch) · In-silico saturation mutagenesis**
  Map every possible variant in one disease enhancer to build a functional fitness map of that regulatory element.

From one broad question to four concrete, scoped projects.

All four share a common toolbox (cheat sheet at the bottom). They differ in which slice of the **variant → molecular effect → gene → tissue → trait** chain they tackle.

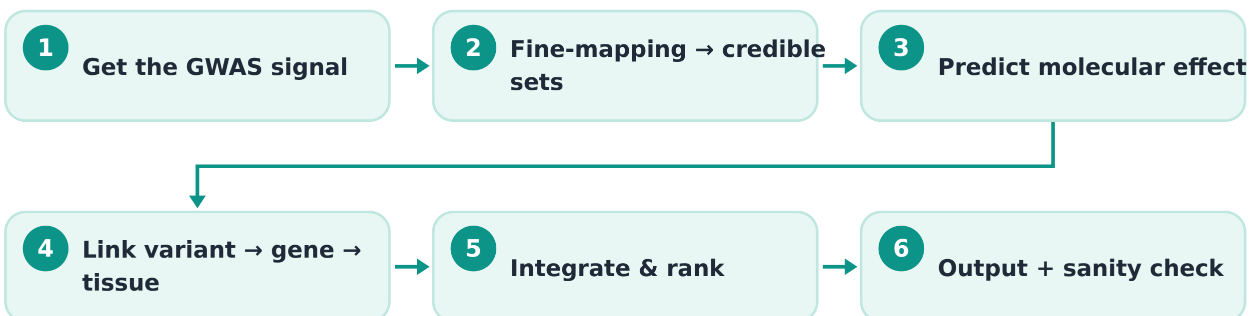
A CANDIDATE A — “GWAS hit to mechanistic hypothesis”

The most direct analog to your PPI → STRING example. It is a clean input-to-output pipeline: feed in a disease, get out a ranked table of “this specific non-coding variant probably causes the signal, here's the mechanism, here's the target gene.”

Narrowed question: *For a single complex disease, can I convert its non-coding GWAS loci into ranked, mechanistically-annotated causal-variant → target-gene hypotheses?*

Pick one disease to scope it. Good choices have well-powered GWAS and decent eQTL coverage: type 2 diabetes, inflammatory bowel disease (IBD), coronary artery disease, or an autoimmune disease. Start with **5–20 loci**, not the whole genome.

PIPELINE



1. **Get the GWAS signal.** Pull summary statistics for your disease from the GWAS Catalog, Open Targets Genetics, or the GWAS Catalog–linked sources. Identify genome-wide significant loci. Output: a list of lead SNPs + surrounding regions (±500 kb windows).
2. **Statistical fine-mapping → credible sets.** Run **SuSiE** (or PolyFun+SuSiE to add functional priors) per locus using the summary stats + an LD reference panel (1000 Genomes or UK Biobank). This narrows each locus from hundreds of correlated variants to a small *credible set* with per-variant posterior inclusion probabilities (PIPs). Output: candidate causal variants per locus, each with a PIP.
3. **Predict the molecular effect of each candidate variant.** Run each credible-set variant through the **AlphaGenome API** (ref vs. alt allele) to get multimodal effect scores: change in chromatin accessibility, TF binding, histone marks, gene expression, and splicing — in disease-relevant tissues/cell types. In parallel, pull the precomputed **CADD v1.7** score and run **SpliceAI/Pangolin** for splicing. Output: a feature matrix (variant × predicted-effect-across-modalities).
4. **Link variant → gene → tissue (colocalization).** For each locus, run **coloc.susie** between the GWAS signal and **GTEx** cis-eQTLs (and sQTLs) across relevant tissues. A high posterior of a shared causal variant (PP4) nominates the target gene and the tissue where

it acts. **Triangulate**, don't trust coloc alone — also record the nearest gene and the gene whose promoter/enhancer AlphaGenome says the variant most affects.

5. **Integrate and rank.** Combine the orthogonal evidence into a single prioritization score per variant: fine-mapping PIP (it's plausibly causal) + AlphaGenome effect magnitude (it does something molecular) + coloc PP4 / target-gene agreement (it points to a gene + tissue). A simple weighted sum or rank-aggregation is fine for a first pass.
6. **Output + sanity check.** Produce a ranked table: *variant* | *locus* | *credible-set PIP* | *predicted mechanism (e.g., "disrupts a GATA motif → loss of enhancer accessibility")* | *target gene* | *tissue* | *supporting evidence*. Validate by checking how many of your top hits recover ClinVar/literature-known mechanisms for that disease (e.g., known regulatory variants), and report concordance.

Expected deliverable: a reproducible notebook + ranked hypothesis table for your chosen disease, plus a short write-up of 2–3 top loci as mechanistic vignettes.

Why it's tractable: every tool is public and most have precomputed data. The hard intellectual work is the *integration logic* in step 5, which is exactly the interesting part.

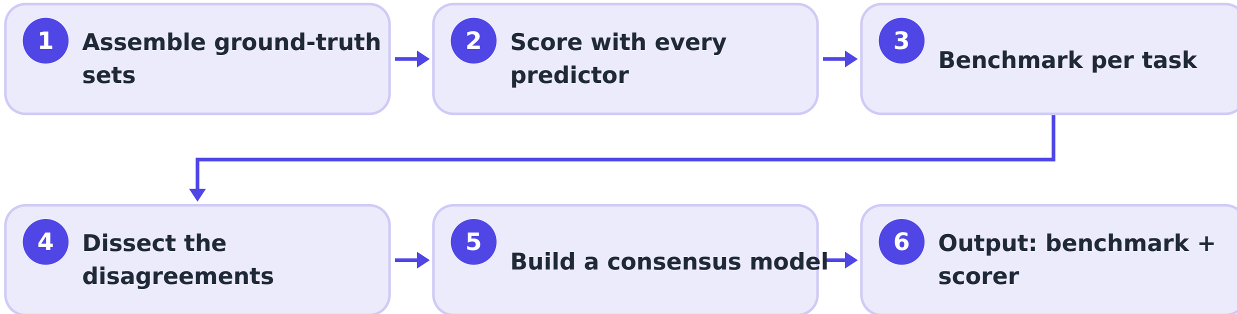
Key pitfall to address head-on: colocalization will sometimes name the wrong gene, and AlphaGenome's effect scores reflect *molecular* effect, not disease causation. Frame outputs as **hypotheses**, and make the triangulation (agreement across independent lines) the headline, not any single score.

B CANDIDATE B — "Do the oracles agree?" (benchmark + meta-predictor)

A methods/evaluation project rather than a discovery project. High value because the field is full of discordant tools and practitioners genuinely don't know which to trust where. Teaches rigorous evaluation, which is a transferable skill.

Narrowed question: *On a gold-standard set of non-coding variants, how do the major predictors compare, where and why do they disagree, and does a simple consensus model beat the best single tool?*

PIPELINE



1. **Assemble ground-truth variant sets.** Pull labeled non-coding variants from several independent “truth” sources so no single bias dominates:
   - **MPRA-measured** variants (functional vs. non-functional regulatory activity) — the cleanest molecular ground truth.
   - **Fine-mapped GTEx eQTLs** (high-PIP causal vs. matched non-causal controls), including *direction* of effect.
   - **ClinVar** pathogenic vs. benign non-coding variants (smaller, clinically labeled).

   Output: a unified table of variants with a label and a task category (regulatory activity / eQTL causality + direction / clinical pathogenicity).
2. **Score every variant with every predictor.** Run **AlphaGenome** (relevant modality heads), precomputed **CADD v1.7**, **SpliceAI + Pangolin** (splicing tasks), and **Borzoi/Enformer** (expression). Standardize outputs into comparable scores. Output: variant × predictor score matrix.
3. **Benchmark per task.** Compute AUROC / AUPRC (and Spearman correlation for quantitative effect sizes) for each predictor on each task. Crucially, also evaluate **direction-of-effect** accuracy on eQTLs, since that's a known weak spot.
4. **Dissect the disagreements.** Stratify performance by variant class: distance to TSS, exonic vs. intronic vs. intergenic, deep-intronic vs. near-splice-site, promoter vs. distal enhancer, common vs. rare. This is where you'll find publishable insight — e.g., confirming/quantifying that splicing tools degrade on deep-intronic variants, or that sequence models drop on distal/individual-level effects.
5. **Build a consensus model.** Train a simple logistic regression or gradient-boosted model that takes all predictor scores as features and outputs a single calibrated probability. Use proper cross-validation with **locus-aware splits** (variants from the same region must not leak across train/test). Test whether the ensemble beats the best single tool, and by how much, per variant class.
6. **Output.** A benchmark report with: per-tool performance tables, a decision guide (“use tool X for variant class Y”), the discordance analysis, and a released consensus score + code.

> **Expected deliverable:** a benchmark + open-source consensus scorer, the kind of artifact other students/labs actually reuse.

Key pitfall to address: data leakage and label bias. MPRA labels reflect *in vitro* activity in one cell type; ClinVar is biased toward canonical/known mechanisms; eQTLs are biased to common variants. State each set's bias and never train and test on overlapping loci.

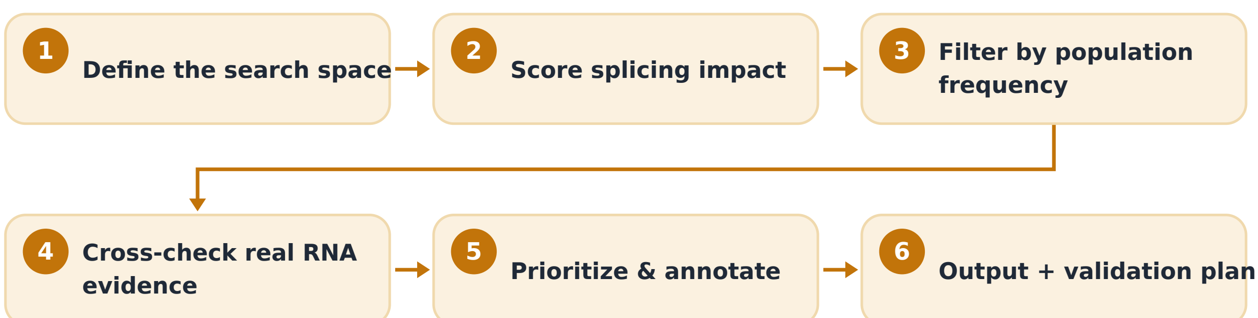
C CANDIDATE C — "Hunting cryptic splice variants in a gene panel"

The narrowest and arguably most feasible. Splicing is the non-coding mechanism with the best predictors and the clearest molecular readout, so the signal-to-effort ratio is high. Strong fit if you want depth over breadth.

Narrowed question: *For a defined disease gene panel, which rare non-coding (deep-intronic / near-splice) variants are predicted to create cryptic splice sites or activate pseudoexons and are therefore candidate-pathogenic?*

Pick a panel where splicing matters and RNA evidence exists — e.g., inherited retinal dystrophy genes (famous for deep-intronic pseudoexon variants like the *CEP290* and *ABCA4* cases), a hereditary cancer panel, or a cardiomyopathy panel. 20–80 genes is plenty.

PIPELINE



1. **Define the search space.** For each gene in the panel, enumerate candidate non-coding positions: introns (especially deep-intronic, >50 bp from exon boundaries), 5'/3' UTRs, and splice-region flanks. Source candidate variants from **gnomAD** (population variants, mostly benign → controls) and **ClinVar VUS / WGS cohorts** (variants of uncertain significance → cases of interest).
2. **Score splicing impact.** Run **SpliceAI**, **Pangolin**, and **AlphaGenome's splicing head** on every candidate (ref vs. alt). Flag variants predicted to (a) create a *de novo* cryptic donor/acceptor, (b) cause exon skipping, or (c) activate a pseudoexon. Use AlphaGenome's splice-junction predictions to get the *predicted aberrant junction*, not just a score.
3. **Filter by population frequency.** Intersect with gnomAD allele frequencies. A strong predicted splice effect that is *absent or ultra-rare* in the population is far more interesting

than a common one (common = probably tolerated). This frequency filter is your cheap proxy for "is it under selection."

4. **Cross-check against real RNA evidence.** Where available, corroborate with **GTEx sQTLs / splice events**, intron-retention signatures, or published RNA-seq for your genes. For top candidates, check whether the predicted cryptic junction is biologically plausible (proper GT-AG dinucleotides, in-frame vs. frameshift consequence, distance creating a viable pseudoexon).
5. **Prioritize and annotate consequence.** Rank candidates by: predicted splice-disruption strength (consensus across the three tools) × rarity × predicted protein consequence (frameshift/PTC introduction is more likely deleterious). Output a per-gene shortlist.
6. **Output + validation plan.** A ranked candidate table per gene with predicted mechanism and aberrant transcript, and a one-page **minigene assay design** for the top 3–5 hits (the standard wet-lab confirmation for splice variants) so the project hands off cleanly to a bench follow-up if one exists.

Expected deliverable: a curated, ranked list of candidate splice-disrupting non-coding variants for a real disease panel — directly useful to a clinical-genetics or research group.

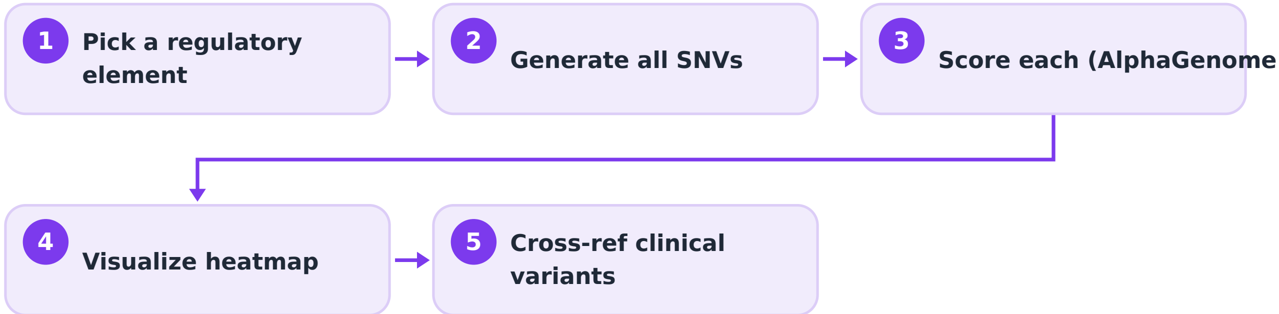
Key pitfall to address: the predictors are weakest exactly in deep-intronic space, so don't over-trust a single high score; require consensus and rationalize each call mechanistically. Note that absence of an effect prediction is *not* evidence of benignity here.

D CANDIDATE D — "In-silico saturation mutagenesis of one regulatory element" (stretch)

A focused single-locus deep dive. Lower breadth, but produces a beautiful, self-contained result and forces you to really understand one model.

Narrowed question: *For one disease-relevant enhancer/promoter, what is the functional consequence of every possible single-nucleotide variant, and which positions/motifs are most sensitive?*

SKETCH



1. Pick a well-studied disease regulatory element (e.g., a known enhancer for an oncogene or a developmental gene).
2. Generate all 3 possible substitutions at every position across the element.
3. Score each with AlphaGenome (and SpliceAI if near a junction) to get a per-base, per-allele effect map.
4. Visualize as a saturation-mutagenesis heatmap; identify the most sensitive positions and overlay known TF motifs (the dips should line up with motif cores — e.g., AlphaGenome reproduces the TAL1/MYB enhancer-creation mechanism, a documented test case you could replicate).
5. Cross-reference any known clinical variants falling in the element against your predicted map.

> **Why it's a nice option:** it mirrors a real published analysis pattern, it's bounded (one element, one weekend of compute), and the figure is striking. Good as a module *inside* Candidate A rather than a standalone if you want one big project.

Shared toolbox / cheat sheet

Layer	Tool / resource	Use	Access
Sequence → multimodal effect	**AlphaGenome**	Predict chromatin, expression, TF binding, splicing from sequence; score variants	Free non-commercial API + Python SDK
Sequence → expression	**Enformer / Borzoi**	Expression/regulatory tracks; baselines	Open source
Splicing	**SpliceAI, Pangolin, AbSplice**	Splice-disruption scores	Open source / web servers
Ensemble deleteriousness	**CADD v1.7**	Genome-wide single score	Precomputed downloads

Fine-mapping	**SuSiE, PolyFun+SuSiE**	GWAS → credible sets + PIPs	R packages
Colocalization	**coloc, coloc.susie, eCAVIAR**	Link signal → gene via eQTL	R packages
GWAS data	**GWAS Catalog, Open Targets Genetics**	Summary stats, loci	Public portals
Molecular QTLs	**GTEx** (eQTL, sQTL)	Tissue-specific gene/splice effects	Public
Population frequency	**gnomAD**	Allele frequencies, constraint	Public
Clinical labels	**ClinVar**	Pathogenic/benign annotations	Public
Functional ground truth	**MPRA datasets** (e.g., MPRAbase, published studies)	Experimentally measured regulatory activity	Public datasets

How to choose

- **Want a discovery pipeline that looks most like the PPI→STRING example → Candidate A**.
- **Stronger in stats/ML and want rigor + a reusable artifact → Candidate B**.
- **Want the most feasible, well-defined, clinically concrete project → Candidate C**.
- **Want something small, elegant, and visual, or a module inside A → Candidate D**.

Two guardrails to bake into whichever you pick

1. **Functional ≠ pathogenic.** A predicted molecular effect is a hypothesis about mechanism, not proof of disease causation — especially for complex traits, where causal signal is the sum of many small effects. Keep these conceptually separate in your outputs.
2. **Triangulate; distrust any single oracle.** Sequence models are strong at “which region/gene is active” but weak at individual-level and distal effects; colocalization is unreliable for naming the target gene on its own. Your project's credibility comes from convergence across independent evidence types, not from one impressive score.